Adaptive Boss Agent - Utility AI Design

Overview

The Adaptive Boss Agent uses a Utility AI architecture to dynamically select actions based on the current game state. Unlike a Finite State Machine (FSM) that follows predefined transitions, Utility AI continuously evaluates available actions and executes the action with the highest utility score. This allows the boss to react intelligently to changing player behavior and combat situations.

Inputs

The boss evaluates the following inputs every decision cycle:

1. Player Weapon Type
2. Player Health Percentage
3. Boss Health Percentage
4. Distance to Player

Available Actions

The boss can perform four tactical actions:

1. Direct Attack

Conditions favoring this action:

- Boss health is high
- Player is close

2. Retreat

Conditions favoring this action:

- Boss health is low
- Immediate danger detected

3. Flank

Conditions favoring this action:

- Player uses ranged weapons
- Player is at medium distance

4. Take Cover

Conditions favoring this action:

- Player uses ranged weapons
- Boss health is moderate or low

Utility AI Architecture

Each action contains one or more considerations.
A consideration:

- Reads a game-state input
- Applies a response curve
- Produces a score between 0 and 1

The scores are combined using a weighted average.
The action with the highest utility score is executed.

Example Scenario

Boss Health: 30%
Player Weapon: Ranged
Distance: Medium

1. Direct Attack
Health Factor = 0.3
Distance Factor = 0.4
Final Score ≈ 0.35

2. Retreat
Formula:
Retreat Score = (1 - Health)^2
Retreat Score = (1 - 0.3)^2
Retreat Score = 0.49

3. Flank
Weapon Factor = 0.9
Distance Factor = 0.6
Final Score ≈ 0.75

4. Take Cover
Weapon Factor = 0.9
Health Factor = 0.7
Final Score ≈ 0.80

Result
Take Cover receives the highest utility score.
The boss chooses to hide because:

- The player has a ranged weapon.
- The boss has low health.
- Attacking directly would be risky.

This demonstrates adaptive and context-aware decision-making.

Why Utility AI Was Chosen

Utility AI was selected because it allows the boss to make dynamic decisions based on the current game state rather than following fixed transitions.
Compared to an FSM, Utility AI provides:

- Greater flexibility
- More adaptive behavior
- Easier scalability for additional actions
- Continuous decision evaluation

The boss agent evaluates multiple tactical options simultaneously and selects the action with the highest utility score.
This creates more believable and intelligent behavior while satisfying the Intelligent Systems requirement for adaptive decision-making.